SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

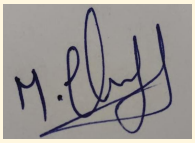
Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – INDIVIDUAL TALK

Course Code:	ITA0301	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 3 – INDIVIDUAL TALK
Weightage:	30 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name:	Madala Virupaksha	**Reg. No.:**	192372205
Section / Batch:	CSE-AI (4th Year)	**Date:**	05/10/2026

Code of Conduct

I Madala Virupaksha certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

- Topic given before the session and students speak on stage.
- Duration: 10 minutes (In class)

Faculty In-charge	Course Coordinator	Program Director

Signature: _____ Date: _____	Signature: _____ Date: _____	Signature: _____ Date: _____

Registration in Cellular Networks

Definition:

Registration (also called Location Update) is the process by which a mobile device informs the cellular network of its current location. This allows the network to know where the user is and deliver calls, SMS, and data correctly.

Steps of Registration

1. The mobile device is switched on or enters a new location area.
2. It sends a Registration/Location Update Request to the nearest base station.
3. The base station forwards the request to the core network.
4. The network authenticates the user using the SIM information.
5. The user's current location is updated in the network database.
6. The network sends a Registration Success message to the mobile device.

Advantages

- Tracks the user's current location.
- Enables incoming calls and messages.
- Supports seamless mobility while moving between cells.
- Reduces unnecessary network searching.

Paging in Cellular Networks

Definition:

Paging is the process used by the cellular network to locate a mobile device when there is an incoming call, SMS, or data packet.

Steps of Paging

1. An incoming call or message arrives for the user.
2. The core network checks the user's last registered location area.
3. A Paging Request is sent to all base stations in that location area.
4. The mobile device receives the paging message.

5. The device responds with a Paging Response.
6. The network establishes the communication connection.

Advantages

- Finds the user quickly.
- Avoids searching every cell in the network.
- Reduces signalling overhead.
- Improves network efficiency.

Difference Between Registration and Paging

Registration	Paging
Initiated by the mobile device	Initiated by the network
Updates the user's location	Locates the user for incoming communication
Occurs when the device is switched on or moves to a new location area	Occurs when there is an incoming call, SMS, or data
Updates the network database	Establishes communication with the mobile device

1. **Geotag Photo:**
2. 